

LAB 2

AGILE BACKLOG CREATION & SPRINT SIMULATION IN JIRA

Problem Statement #09

FACULTY RESEARCH GRANT & PUBLICATION TRACKER

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Objective

Use Jira to turn the Lab 1 functional requirements into Agile backlog items, simulate a sprint, and analyse progress.

1. Set Up Your Jira Project

Project Type: Company-managed Scrum project

Project Name: Faculty Research Grant & Publication Tracker

Project Key: FRGPT

Sprint model: Scrum

Sprint duration required by the handout: 1 week

2. Create Epics

Epic 1: Research Grant & Financial Management

Description: Enable Faculty Researchers to create and maintain grant records, record research expenses, validate expenses against approved funding limits, and view remaining grant balances.

Mapped Lab 1 requirements: FR-001, FR-002

Epic 2: Publication & Co-author Management

Description: Enable Faculty Researchers to record publications, maintain indexing/citation information, and submit and track co-author approvals.

Mapped Lab 1 requirements: FR-003, FR-004

Epic 3: Research Analytics & Performance Monitoring

Description: Enable the Research Dean to monitor fund utilization and research/publication performance through analytics and metrics.

Mapped Lab 1 requirements: FR-005

3. Write User Stories

The user stories follow the required format: As a [user role], I want [goal], So that [benefit].

Story 1.1: Manage Research Grant Records

As a	I want to	So that
As a Faculty Researcher	I want to create, update, and view research grant records including allocation and approved funding limits	So that sponsored research funding is maintained accurately.

Priority: High Story Points: 5 Requirement: FR-001

Acceptance Criteria: A researcher can create/update a grant record and the system displays the correct allocated amount and funding limit.

Story 1.2: Record Research Expenses

As a	I want to	So that
As a Faculty Researcher	I want to record equipment procurement and other research expenses against an active grant	So that grant expenditure is tracked centrally.

Priority: High Story Points: 5 Requirement: FR-002

Acceptance Criteria: A valid expense updates the remaining balance accurately; an invalid or overdraft expense is rejected.

Story 1.3: Validate Expense Against Grant Limit

As a	I want to	So that
As a Faculty Researcher	I want the system to validate an expense against the remaining grant balance	So that overspending is prevented.

Priority: High Story Points: 3 Requirement: FR-002

Acceptance Criteria: An expense within the available balance is accepted; an expense exceeding the available balance is rejected.

Story 1.4: View Remaining Grant Balance

As a	I want to	So that
As a Faculty Researcher	I want to view the updated grant balance after valid expenses	So that I know how much funding remains available.

Priority: High Story Points: 3 Requirement: FR-002

Acceptance Criteria: The remaining balance is recalculated and displayed after a valid expense.

Story 2.1: Manage Publication Records

As a	I want to	So that
As a Faculty Researcher	I want to add, update, and view publication records	So that my research output is maintained in one place.

Priority: High Story Points: 5 Requirement: FR-003

Acceptance Criteria: A publication can be recorded and its indexing/citation information can be stored and updated.

Story 2.2: Submit Publication for Co-author Approval

As a	I want to	So that
As a Faculty Researcher	I want to submit a publication for co-author approval	So that the publication can follow a structured approval process.

Priority: Medium Story Points: 3 Requirement: FR-004

Acceptance Criteria: A publication can be submitted for co-author approval.

Story 2.3: Track Co-author Approval Status

As a	I want to	So that
As a Faculty Researcher	I want to track co-author approval status	So that I know whether a publication has been approved.

Priority: Medium Story Points: 3 Requirement: FR-004

Acceptance Criteria: Approval status changes according to approval actions and is visible.

Story 2.4: Maintain Publication Metrics

As a	I want to	So that
As a Faculty Researcher	I want to maintain indexing status and citation information	So that publication impact can be tracked.

Priority: Medium Story Points: 5 Requirement: FR-003

Acceptance Criteria: Indexing and citation details can be stored and updated.

Story 3.1: View Fund Utilization Analytics

As a	I want to	So that
As a Research Dean	I want to view grant fund utilization and remaining funding	So that I can monitor sponsored-fund usage.

Priority: Medium Story Points: 5 Requirement: FR-005

Acceptance Criteria: The Research Dean can view grant utilization and remaining funding.

Story 3.2: View Research Performance Metrics

As a	I want to	So that
As a Research Dean	I want to view research and publication performance information	So that I can monitor faculty research performance.

Priority: Medium Story Points: 5 Requirement: FR-005

Acceptance Criteria: The Research Dean can view research and publication performance information.

4. Prioritize the Backlog

Priority is assigned according to user impact and the dependency of core grant and expense functions on later publication and analytics workflows.

Rank	User Story	Priority	Story Points	Requirement
1	Story 1.1: Manage Research Grant Records	High	5	FR-001
2	Story 1.2: Record Research Expenses	High	5	FR-002
3	Story 1.3: Validate Expense Against Grant Limit	High	3	FR-002
4	Story 1.4: View Remaining Grant Balance	High	3	FR-002
5	Story 2.1: Manage Publication Records	High	5	FR-003

6	Story 2.2: Submit Publication for Co-author Approval	Medium	3	FR-004
7	Story 2.3: Track Co-author Approval Status	Medium	3	FR-004
8	Story 2.4: Maintain Publication Metrics	Medium	5	FR-003
9	Story 3.1: View Fund Utilization Analytics	Medium	5	FR-005
10	Story 3.2: View Research Performance Metrics	Medium	5	FR-005

5. Assign Story Points

The handout specifies Fibonacci-style estimation. The proposed estimates are based on relative complexity, amount of work and uncertainty.

Story	Points	Planning Poker / Estimation Rationale
Story 1.1	5	Grant record creation/update and funding fields
Story 1.2	5	Expense recording and grant linkage
Story 1.3	3	Validation against remaining balance
Story 1.4	3	Balance calculation and display
Story 2.1	5	Publication record management
Story 2.2	3	Approval submission workflow
Story 2.3	3	Approval status tracking
Story 2.4	5	Indexing and citation metrics
Story 3.1	5	Fund utilization analytics
Story 3.2	5	Research/publication performance analytics

Total backlog estimate: 42 story points.

6. Run Your Sprint

Sprint 1 Goal: Complete the core grant, expense and publication-record workflows and the co-author approval submission workflow.

Sprint 1 Story	Priority	Points	Final Simulated Status
Story 1.1: Manage Research Grant Records	High	5	Done
Story 1.2: Record Research Expenses	High	5	Done
Story 1.3: Validate Expense Against Grant Limit	High	3	Done
Story 1.4: View Remaining Grant Balance	High	3	Done
Story 2.1: Manage Publication Records	High	5	Done
Story 2.2: Submit Publication for Co-author Approval	Medium	3	Done

Sprint 1 committed scope: 24 story points. The simulated sprint moves the selected stories from To Do → In Progress → Done.

Sprint 1 Daily Simulation

Time	Remaining Story Points	Progress
Start	24	All selected stories in To Do
Day 1	21	Grant record workflow completed
Day 2	17	Expense recording workflow completed
Day 3	12	Expense validation and balance update completed
Day 4	8	Publication record management completed
Day 5	5	Co-author approval submission completed
Day 6	2	Final verification
Day 7	0	Sprint goal achieved

Screenshot — Jira Backlog with Epics and User Stories

FRG-PT

Faculty Research Grant & Publication Tracker

Summary

Backlog

Active sprints

Reports

Issues

Project settings

Backlog

Faculty Research Grant & Publication Tracker

Create

Epics

FRGPT-1

Research Grant & Financial Management

Requirements: FR-001, FR-002

FRGPT-2

Publication & Co-author Management

Requirements: FR-003, FR-004

FRGPT-3

Research Analytics & Performance Monitoring

Requirements: FR-005

Issues / User Stories

Type	Key	Summary	Priority	Story Points	Status
	FRGPT-11	Manage Research Grant Records	High	5	To Do
	FRGPT-12	Record Research Expenses	High	5	To Do
	FRGPT-13	Validate Expense Against Grant Limit	High	3	To Do
	FRGPT-14	View Remaining Grant Balance	High	3	To Do
	FRGPT-15	Manage Publication Records	High	5	To Do
	FRGPT-16	Submit Publication for Co-author Approval	Medium	3	To Do
	FRGPT-17	Track Co-author Approval Status	Medium	3	Backlog
	FRGPT-18	Maintain Publication Metrics	Medium	5	Backlog

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Screenshot — Story Point Assignments and Priorities

FRG-PT

Faculty Research Grant & Publication Tracker

Summary

Backlog

Active sprints

Reports

Issues

Project settings

Backlog

Story points and priorities

Sprint 1

24 story points committed

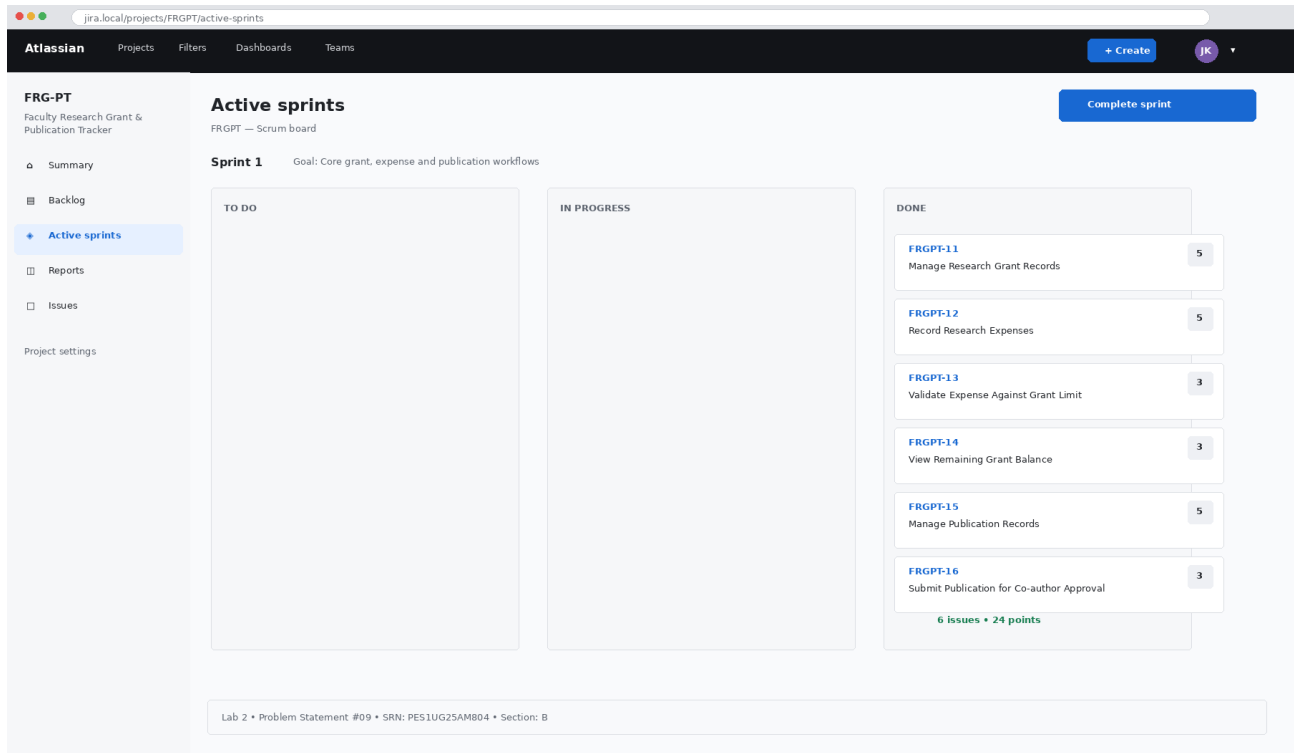
Estimation: Fibonacci

Priority: High → Medium

Key	Summary	Epic	Priority	Story points	Sprint
FRGPT-11	Manage Research Grant Records	FRGPT-1	High	5	Sprint 1
FRGPT-12	Record Research Expenses	FRGPT-1	High	5	Sprint 1
FRGPT-13	Validate Expense Against Grant Limit	FRGPT-1	High	3	Sprint 1
FRGPT-14	View Remaining Grant Balance	FRGPT-1	High	3	Sprint 1
FRGPT-15	Manage Publication Records	FRGPT-2	High	5	Sprint 1
FRGPT-16	Submit Publication for Co-author Approval	FRGPT-2	Medium	3	Sprint 1
FRGPT-17	Track Co-author Approval Status	FRGPT-2	Medium	3	Backlog
FRGPT-18	Maintain Publication Metrics	FRGPT-2	Medium	5	Backlog
FRGPT-19	View Fund Utilization Analytics	FRGPT-3	Medium	5	Backlog
FRGPT-20	View Research Performance Metrics	FRGPT-3	Medium	5	Backlog

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Screenshot — Sprint 1 Active Sprint Board



7. Finish 2 Sprints

Sprint 2 completes the remaining publication metrics, approval tracking, and Research Dean analytics.

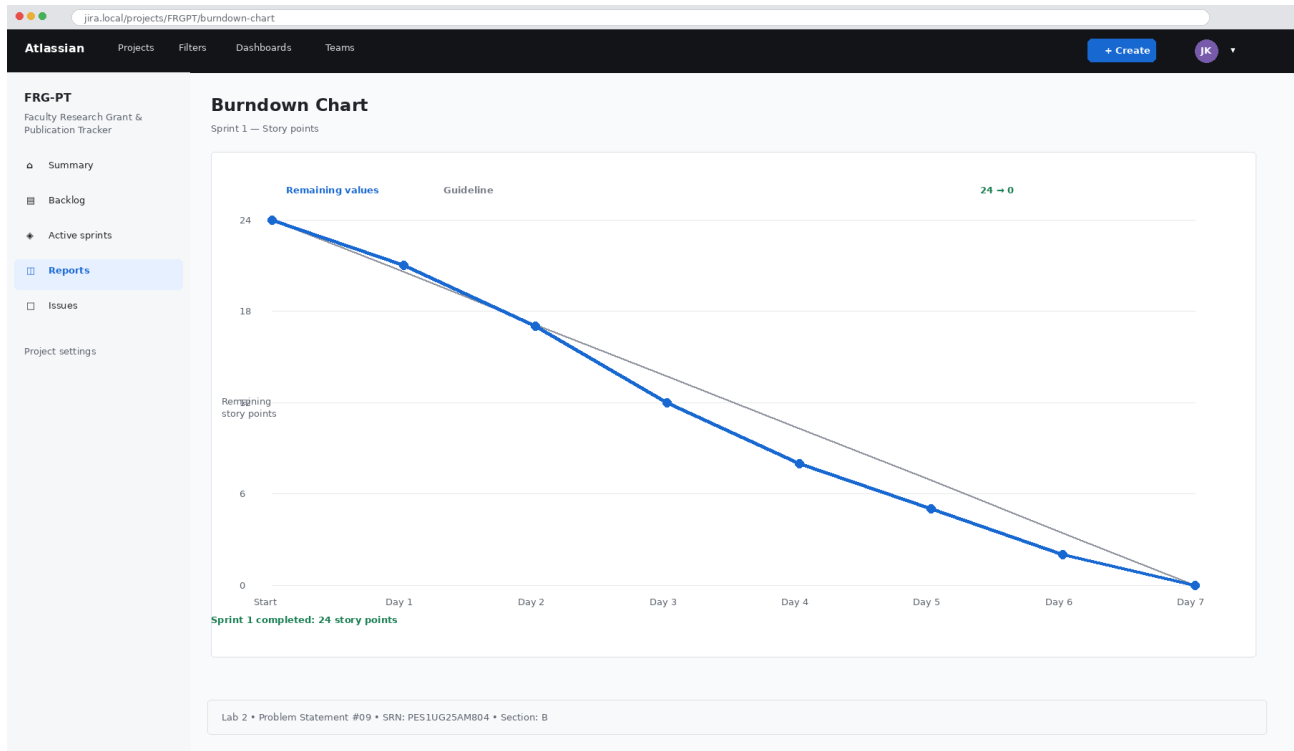
Sprint 2 Story	Priority	Points	Final Simulated Status
Story 2.3: Track Co-author Approval Status	Medium	3	Done
Story 2.4: Maintain Publication Metrics	Medium	5	Done
Story 3.1: View Fund Utilization Analytics	Medium	5	Done
Story 3.2: View Research Performance Metrics	Medium	5	Done

Sprint 2 committed scope: 18 story points. The remaining functional backlog is completed in the simulation.

8. Generate a Burndown Chart

The Lab 2 handout asks the team to observe remaining work across the sprint and reflect on delays, early completions or inconsistencies.

Screenshot — Jira Burndown Chart



Interpretation: The simulated remaining-work line decreases from 24 points to 0 by Day 7. This indicates that the selected Sprint 1 scope was achievable within the simulated sprint capacity.

Reflection Questions

1. Did your estimations reflect the actual effort?

The estimates were reasonably aligned with the simulated effort. Three-point stories represent focused validation or workflow functions, while five-point stories represent broader record-management or analytics work. Because this is a simulation, actual development effort cannot be measured from the simulation alone.

2. Was your backlog well-prioritized?

Yes. Core grant and expense functions were given High priority because they support funding accuracy and prevention of overspending. Publication management was next, followed by approval tracking and Research Dean analytics.

3. How did your simulated sprint align with your plan?

Sprint 1 selected the core grant, expense and publication workflows and completed all 24 committed points in the simulation. Sprint 2 completed the remaining four stories and 18 points.

4. What insights did the burndown chart give about your team's capacity?

The burndown shows a steady reduction in remaining work and reaches zero by the end of the sprint. This suggests that the simulated Sprint 1 commitment was achievable. More real sprint history would be needed to establish reliable team velocity.

Deliverables

- 1. EPICs
- 2. User Stories

- 3. Priority and Story Point Assignments
- 4. Sprint 1 Active Sprint View
- 5. Sprint 2 Completion
- 6. Burndown Chart
- 7. Reflection Questions